

THE CHEMISTRY OF GUNPOWDER

Average thing that happens in gunpowders:



Gunpowder is a heterogeneous solid mixture that undergoes rapid combustion, not detonation.

Key characteristics:

- Solid reactants
- Fast reaction rate
- Produces hot gases
- Energy released as heat and pressure



I WANT YOU
TO BLOW THIS UP
(VOTES)

Energy Changes (Thermochemistry)

- Energy stored in chemical bonds
- Bonds break → new bonds form
- Excess energy released as heat (exothermic reaction)

The reaction has:

- Low activation energy once ignited
- Very high energy output as heat over a short time

According to the Ideal Gas Law ($PV=nRT$):

Increasing temperature increases gas pressure.

Hot gases expand rapidly, pushing the bullet out of the gun at a super high speed.